

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2002

MATHEMATICS EXTENSION 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is supplied at the back of this paper
- Start each question on a new page
- Attempt all Questions 1 – 8
- All questions are of equal value

Name : _____

Class : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL

Question 1 15 marks

Marks

a) Find $\int \sin^3 x \, dx$ 2

b) Find $\int \frac{dx}{\sqrt{2x - x^2}}$ 2

c) Use partial fractions to find $\int \frac{5}{(x-3)(2x-1)} \, dx$ 3

d) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ using the substitution $t = \tan \frac{x}{2}$ 4

e) Find $\int e^x \sin x \, dx$ 4

Question 2 15 marks (Use a new page)

Marks

a) Express $\frac{\overline{2+3i}}{1+i}$ in the form $x+iy$. 2

b) Given $\omega = -1+i\sqrt{3}$ find i) $|\omega|$ 1

ii) $\arg(\omega)$ 1

iii) $\omega^5 + 16\omega$ in the form $x+iy$ 2

c) On an Argand diagram shade the region specified by

i) $|z-1| \leq |z-i|$ 1

ii) $\operatorname{Re}\left(\frac{2}{z}\right) \leq 1$ 3

d) One of the square roots of $a+3i$ is equal to $3+bi$
where a and b are real. Find the value of a and b . 2

e) Sketch the region where both the following inequalities hold. 3

$$|z-3| \leq 3 \quad \text{and} \quad 0 \leq \arg(z) \leq \frac{\pi}{2}$$

Question 3 15 marks (Use a new page)

Marks

a) Consider the function $f(x) = \frac{x}{\ln x}$

i) State the natural domain of $f(x)$ 1

ii) Show that the curve $y = f(x)$ 2

has a minimum turning point at the point (e, e) .

iii) Find the point of inflexion of the curve $y = f(x)$. 1

iv) Sketch the curve $y = f(x)$ showing the important features. 2

b) $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ where $p, q > 0$ are two distinct points on

the hyperbola H with equation $xy = c^2$.

i) Show that the equation of the tangent to H at $P(cp, \frac{c}{p})$ 2

is given by $x + p^2y = 2cp$.

ii) The tangents to H at $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ meet at T . 2

Find the coordinates of T .

iii) Find the equation of the chord PQ . 2

iv) Given that the chord PQ passes through the point $(2c, 0)$ 1

find a relationship between the parameters p and q .

v) Find the equation of the locus of T as P and Q move about H 2

according to the restriction in part iv).

Give a complete description of this locus.

Question 4 15 marks (Use a new page)

Marks

- a) The area bounded by $y = \frac{1}{x+1}$, the x axis

4

and the lines $x = 0$ and $x = 2$ is rotated about the line $x = 2$.

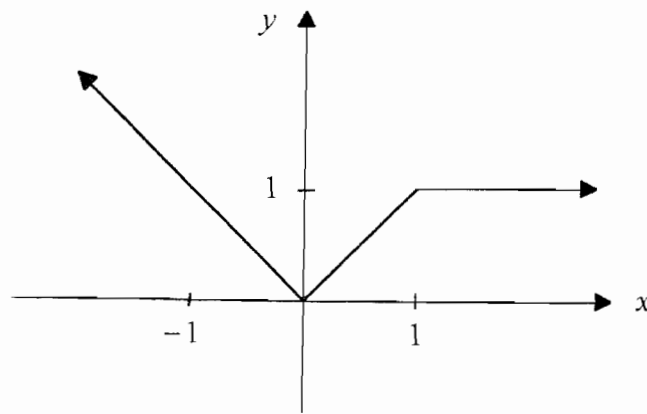
Use cylindrical shells to find the volume of the solid of revolution formed.

- b) Find the equation of the ellipse with centre the origin,

3

which has a focus at $(2,0)$ and the corresponding directrix is $x = 4$.

- c)



The diagram shows the graph of the function $y = f(x)$.

Draw separate sketches of the following

i) $y = f(-x)$

1

ii) $y = f(|x|)$

1

iii) $y = \ln(f(x))$

2

iv) $x = f(y)$

1

- d) If ω is a complex root of $z^3 = 1$ find the value of

i) $1 + \omega + \omega^2$

1

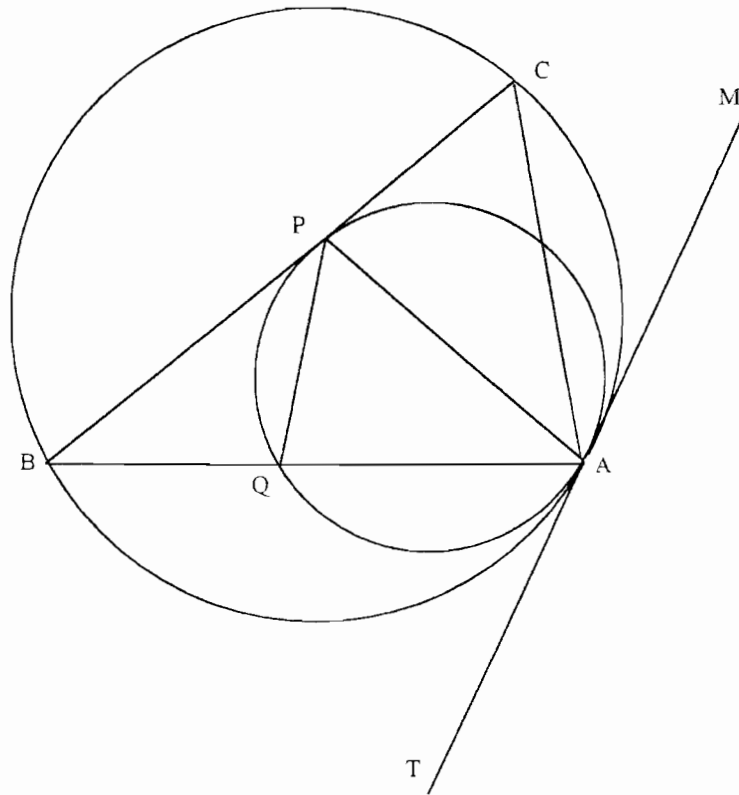
ii) $\frac{1}{3 + 5\omega + 3\omega^2} + \frac{1}{7 + 7\omega + 9\omega^2}$

2

Question 5 15 marks (Use a new page)

Marks

a)



Two circles touch internally at A . MT is a common tangent.

A tangent to the inner circle at P cuts the outer circle at B and C .

The interval AB cuts the inner circle at Q .

The intervals PA , CA and PQ have been drawn.

Neatly draw the diagram on your answer sheet.

Prove that PA bisects $\angle BAC$.

Question 5 (Continued)

Marks

b) The equation $2x^4 - 3x^2 - 2x + k = 0$ has a triple root.

4

Find the value of k .

c) Solve $x^4 - 6x^3 + 15x^2 - 18x + 10 = 0$ over the complex field

4

given that $1 + i$ is a solution.

d) On an Argand diagram $\triangle POQ$ is a right angled isosceles triangle.

O is the origin, P lies in the first quadrant and Q lies in the second.

$$\angle POQ = 90^\circ.$$

Given that OP represents the complex number $a + ib$,

Write down the complex number represented by

i) OQ

1

ii) PQ

1

Question 6 15 marks (Use a new page)

Marks

a) The function f is given by

$$f(x) = e^{\frac{x}{1+kx}} \text{ where } k \text{ is a positive constant.}$$

i) Given that the tangent to $y = f(x)$ at the point $(a, f(a))$

3

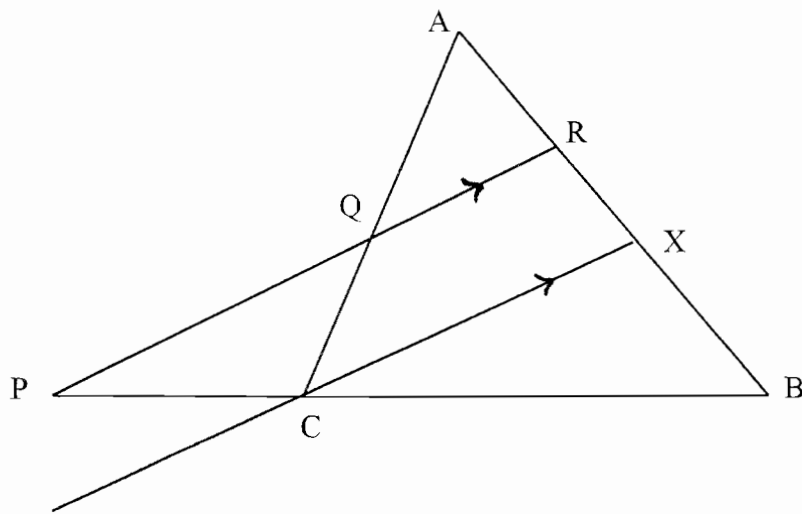
passes through the origin show that

$$k^2 a^2 + (2k - 1)a + 1 = 0.$$

ii) Deduce that there is no such tangent as in part i) if $4k > 1$.

2

b)



ABC is a triangle. The line RQ produced meets BC produced at P .

CX is drawn parallel to QR .

Show that $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{AR}{RB} = 1$

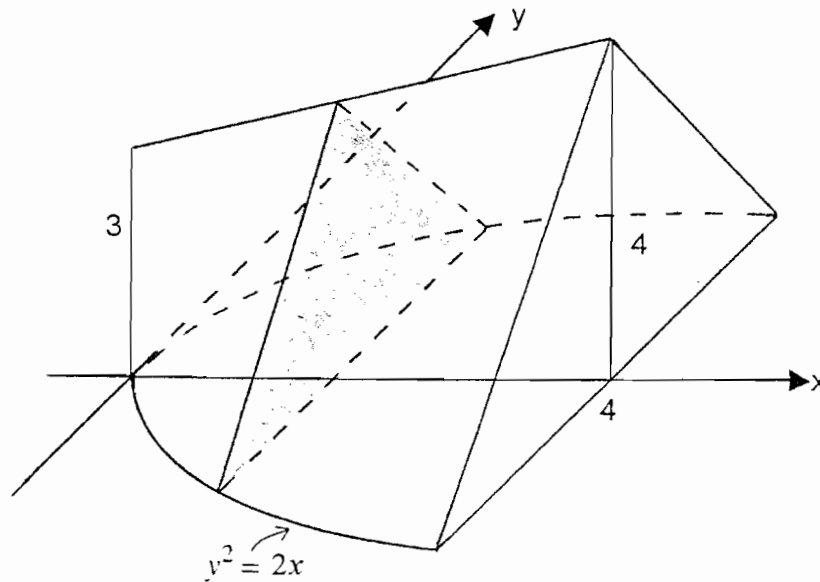
4

(This is known as Menelaus' Theorem.)

Question 6 (Continued)

Marks

c)



The base of the above solid is the area enclosed by $y^2 = 2x$ and $x = 4$.

Each cross section of the solid by planes perpendicular to the x axis is an isosceles triangle with the equal sides meeting above the base of the solid. A typical cross section of thickness Δx has been shaded.

The perpendicular height of the solid at $x = 4$ is 4 and the perpendicular height of the solid at $x = 0$ is 3.

A straight line joins the vertex of the triangle at $x = 4$ to the point 3 above $x = 0$.

i) Show that the perpendicular height of the solid as a function of x

2

is given by $\frac{1}{4}x + 3$

ii) Find the volume of the solid.

4

Question 7 15 marks (Use a new page)

Marks

a) Given that $I_n = \int x^n e^{2x} dx$

i) Show that $I_n = \frac{x^n e^{2x}}{2} - \frac{n}{2} I_{n-1}$ 3

ii) Use the above result to find $\int x^2 e^{2x} dx$. 2

b) A particle of mass $2kg$ is propelled from the origin along the

x axis with an initial velocity of $Q m/s$.

The only forces acting on the body in the direction of the x axis are

friction which is a constant 16 Newtons and air resistance

which equals v^2 Newtons where v is the velocity of the particle

t seconds after leaving the origin.

i) Explain why $\frac{dv}{dt} = -8 - \frac{1}{2}v^2$. 1

ii) Show that $t = \frac{1}{2} \tan^{-1}(\frac{4Q - 4v}{16 + Qv})$ 5

iii) By using $\frac{dv}{dt} = v \frac{dv}{dx}$ 4

find an expression for v in terms of x .

Question 8 15 marks (Use a new page)

Marks

- a) Solve the equation

4

$$\sin(x + 10^\circ) = \cos(4x) \quad \text{for } 0^\circ \leq x \leq 180^\circ.$$

- b) Use Mathematical Induction to prove that

4

$$\tan \theta + 2 \tan 2\theta + \dots + 2^{n-1} \tan(2^{n-1} \theta) = \cot \theta - 2^n \cot(2^n \theta)$$

for all positive integers n .

- c) The function $f(x)$ is given, for $x > 0$, by

$$f(x) = 2 \log_e x - \frac{x^2 - 1}{x}.$$

- i) Show that the only zero of $f(x)$ occurs at $x = 1$.

3

Justify your answer.

- ii) Let $g(x) = \frac{x \log_e x}{x^2 - 1}$, for $x > 0$ and $x \neq 1$.

4

Show that $0 < g(x) < \frac{1}{2}$.

End of Test.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Teacher's Name:

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Q1

$$\begin{aligned} \int \sin^3 x \, dx &= \int \sin x (1 - \cos^2 x) \, dx \quad \text{let } u = \cos x \\ &= \int u^2 - 1 \, du \quad du = -\sin x \, dx \\ &= \int u^2 - 1 \, du \\ &= \frac{1}{3} \cos^3 x - \cos x + C \end{aligned}$$

$$\begin{aligned} \int \frac{dx}{\sqrt{2x-x^2}} &= \int \frac{dx}{\sqrt{1-(x-1)^2}} \\ &= \sin^{-1}(x-1) + C \\ &= \sin^{-1}(1-x) + C \end{aligned}$$

$$\frac{S}{(x-3)(2x-1)} = \frac{A}{x-3} + \frac{B}{2x-1}$$

$$\begin{aligned} x=3 & \Rightarrow S=3A \Rightarrow A=1 \\ x=\frac{1}{2} & \Rightarrow S=-2B \Rightarrow B=-2 \end{aligned}$$

$$\begin{aligned} \int \frac{S}{(x-3)(2x-1)} \, dx &= \int \frac{1}{x-3} - \frac{2}{2x-1} \, dx \\ &= \ln|x-3| - \ln|2x-1| + C \\ &= \ln\left(\frac{x-3}{2x-1}\right) + C \end{aligned}$$

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$$\begin{aligned} \int_0^{\frac{\pi}{6}} \frac{dx}{2+\cos x} &= \int_0^{\frac{\pi}{6}} \frac{2dx}{3+4\cos^2 x} \\ &= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} \right]_0^{\frac{\pi}{6}} \\ &= \frac{2}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right) \\ &= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6} \\ &= \frac{\pi}{3\sqrt{3}} \end{aligned}$$

$$\begin{aligned} \int e^{2x} \sin x \, dx &= -e^{2x} \cos x + \int e^{2x} \cos x \, dx \\ &= -e^{2x} \cos x + \left[e^{2x} \sin x - \int e^{2x} \sin x \, dx \right] \end{aligned}$$

$$\int e^{2x} \sin x \, dx = -e^{2x} \cos x + e^{2x} \sin x - \int e^{2x} \sin x \, dx$$

$$2 \int e^{2x} \sin x \, dx = e^{2x} (\sin x - \cos x)$$

$$\int e^{2x} \sin x \, dx = \frac{e^{2x}}{2} (\sin x - \cos x) + C$$

Teacher's Name:

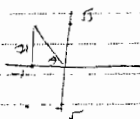
Student's Name/N°:

Q2

$$\begin{aligned} \frac{2+3i}{1+i} &= \frac{2+3i}{1+i} \cdot \frac{1-i}{1-i} \\ &= \frac{2-2i-3i-3}{1-i^2} \\ &= \frac{-1-5i}{2} \\ &= -\frac{1}{2} - \frac{5}{2}i \end{aligned}$$

$$|w| = \sqrt{(-1)^2 + (5)^2}$$

$$\arg(w) = \frac{5\pi}{2}$$



$$w^2 + 16w$$

$$= \left(2\sqrt{6}\frac{\sqrt{3}}{2}\right)^2 + 16\left(2\sqrt{6}\frac{\sqrt{3}}{2}\right)$$

$$= 32\sqrt{6}\frac{\sqrt{3}}{2} + 32\sqrt{6}\frac{\sqrt{3}}{2}$$

$$= 32\left(\sqrt{6}\frac{\sqrt{3}}{2} + \sqrt{6}\frac{\sqrt{3}}{2}\right)$$

$$= 32\left(\sqrt{6}\frac{\sqrt{3}}{2} + \sqrt{6}\frac{\sqrt{3}}{2}\right)$$

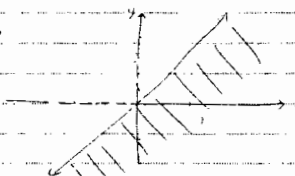
$$= 64\sqrt{6}\frac{\sqrt{3}}{2}$$

$$= 32\sqrt{2}$$

Teacher's Name:

Student's Name/N°:

Q1



$$\begin{aligned} \operatorname{Re}\left(\frac{z}{\delta}\right) &\leq 1 \\ \frac{2}{\delta} &= \frac{2}{x+iy} \cdot \frac{x-iy}{x-iy} \\ &= \frac{2(x-iy)}{x^2+y^2} \end{aligned}$$

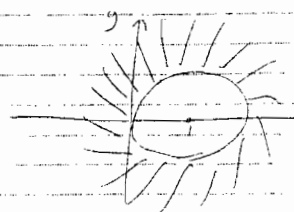
$$\operatorname{Re}\left(\frac{z}{\delta}\right) \leq 1$$

$$\frac{2x}{x^2+y^2} \leq 1$$

$$2x \leq x^2+y^2$$

$$x^2-2x+y^2 \geq 0$$

$$(x-1)^2+y^2 \geq 1$$



$$\sqrt{a+bi} = 3+6i$$

$$a+bi = (3+6i)^2$$

$$= 9-36+36bi$$

$$\text{equating real + imaginary parts}$$

$$9-36=a \quad \text{and} \quad 36=b$$

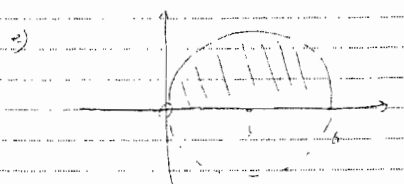
$$\Rightarrow b=36$$

$$9-\frac{36}{2}=a$$

$$a=-9$$

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Q3

g.f. $x > 0, x \neq 1$

$$i) y = \frac{x}{\ln x}$$

$$y' = \frac{(\ln x)(1) - x(1/x)}{(\ln x)^2}$$

$$= \frac{\ln x - 1}{(\ln x)^2}$$

st. pt where $y' = 0$

$$\ln x - 1 = 0$$

$$(\ln x) = 1$$

$$x = e$$

$$\text{when } x = e, y = 1$$

$$y'' = \frac{(\ln x)^2 \left(\frac{1}{x}\right) - (\ln x - 1)(2 \ln x)}{(\ln x)^4}$$

$$= \frac{2 - \ln x}{x(\ln x)^3}$$

$$\text{test st. pt: when } x = e, y''(e) = \frac{2 - \ln e}{e(\ln e)^3}$$

$$= \frac{1}{e}$$

$$\therefore \text{min at } (e, 1)$$

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$$ii) \text{ inflection at } y'' = 0$$

$$2 - \ln x = 0$$

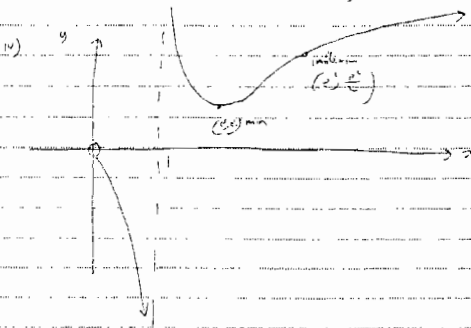
$$\ln x = 2$$

$$x = e^2$$

$$\text{when } x = e^2, y = \frac{e^2}{\ln e^2}$$

$$= \frac{e^2}{2}$$

$$\therefore \text{pt of inflection } \left(e^2, \frac{e^2}{2}\right)$$



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Q3 b.

$$i) y = e^x x^{-1}$$

$$\frac{dy}{dx} = e^x x^{-2}$$

$$\text{when } x = p$$

$$m_p = \frac{1}{e^p p^2}$$

$$= \frac{1}{p^2}$$

equation of tangent

$$y - \frac{1}{p} = \frac{1}{p^2}(x - p)$$

$$p^2 y - p = -x + p$$

$$x + p^2 y = 2p$$

$$\text{if } x + p^2 y = 2p \text{ and } x + p^2 y = 2p \text{ simultaneously}$$

$$x + p^2 y = 2p$$

$$p^2 y - p = -x + p$$

$$y = \frac{2p - x}{p^2}$$

$$= \frac{2p}{p^2}$$

$$x + p^2 \left(\frac{2p}{p^2}\right) = 2p$$

$$x + 2p = 2p$$

$$x = 0$$

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$$ii) m_{pq} = \frac{1}{p^2}$$

$$= \frac{1}{p^2}$$

equation

$$y - \frac{1}{p} = \frac{1}{p^2}(x - p)$$

$$p^2 y - p = -x + p$$

$$x + p^2 y = 2p$$

$$iv) \text{ Sub } (x, y) \text{ into above}$$

$$2x + 0 = 2(p^2 y)$$

$$p^2 y = p$$

$$v) T \left(\frac{2p}{p^2}, \frac{2p}{p^2} \right)$$

$$x = \frac{2p}{p^2}$$

$$y = \frac{2p}{p^2}$$

$$\text{but } p^2 y = p$$

$$\therefore \text{but tangent must meet curve}$$

$$\text{at } (p, \frac{1}{p}) \text{ in 1st quadrant}$$

$$\therefore 0 < p < e$$

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Q.4

 $\Delta V = 2\pi \text{ radius} \times \text{height} \times \text{thickness}$

$$= 2\pi (2-x) \times y \times dx$$

$$= 2\pi (2-x) \times \frac{1}{x+1} dx$$

$$\therefore V = \int_0^2 2\pi (2-x) \times \frac{1}{x+1} dx$$

$$= 2\pi \int_0^2 \frac{2-x}{x+1} dx$$

$$= 2\pi \int_0^2 \frac{2}{x+1} - \frac{x}{x+1} dx$$

$$= 2\pi \int_0^2 \frac{2}{x+1} - \frac{x+1-1}{x+1} dx$$

$$= 2\pi \left[2\ln(x+1) - x \right]_0^2$$

$$= 2\pi \left[(2\ln 3 - 2) - (2\ln 1 - 0) \right]$$

$$= 2\pi \left[2\ln 3 - 2 \right] \text{ cm}^2$$

$$b) \quad a^2 = 2 \quad \frac{a}{2} = 4$$

$$\Rightarrow a = 4\sqrt{2}$$

$$\therefore 4x(x) = 2$$

$$2x^2 = 2$$

$$x^2 = 1$$

$$x = \pm 1$$

$$a = \frac{4}{\sqrt{2}}$$

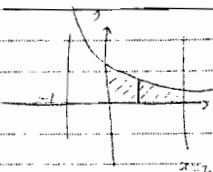
$$= 2\sqrt{2}$$

$$b^2 = a^2(1-e^2)$$

$$= 5(1-\frac{1}{2})$$

$$= \frac{5}{2}$$

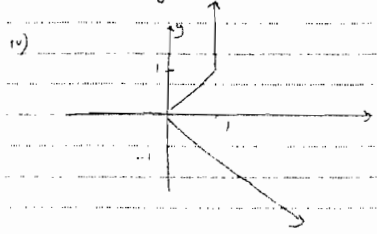
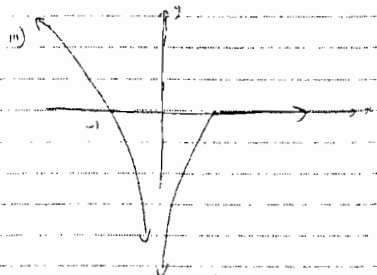
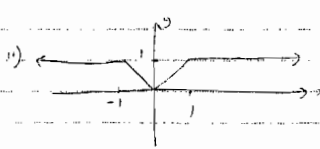
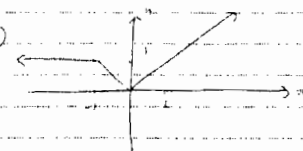
$$\therefore \text{equation } \frac{x^2}{8} + \frac{y^2}{4} = 1$$



Teacher's Name:

Student's Name/Nº:

Q.5



Teacher's Name:

Student's Name/Nº:

(2.4.d) i) If w is a root so is w^2

$$w^2 = 1 \quad \therefore (w^2)^2 = (w^2)^2 = 1^2 = 1$$

$$\therefore w^2 \text{ is a root}$$

$$1 + w + w^2 = 0 \quad (\text{sum of roots})$$

$$ii) \quad \frac{1}{5+5w+3w^2} + \frac{1}{7+7w+9w^2}$$

$$= \frac{1}{3(2w+3w^2+2w)} + \frac{1}{7(7w+7w^2+7w^2)}$$

$$= \frac{1}{2w} + \frac{1}{2w^2}$$

$$= \frac{1}{2} \left[\frac{w^2+1}{w^2} \right]$$

$$= \frac{1}{2} \left[\frac{w^2+1}{w^2} \right] \quad (w^2=1, 1+w+w^2=0)$$

$$= \frac{1}{2} \left[\frac{1}{1} \right]$$

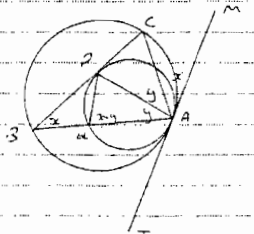
$$= \frac{1}{2}$$

Teacher's Name:

Student's Name/Nº:

Q.5

a)



$$\therefore \angle MAC = x, \angle PAC = y$$

$$\therefore \angle ABC = x \quad (\text{angle in alternate segment})$$

$$\angle PAA = 2xy \quad (\text{angle in alternate segment})$$

$$\therefore \angle BPA = y \quad (\text{ext. angle of } \triangle \text{ equals sum of 2 remote interior angles})$$

$$\therefore \angle PAA = y \quad (\text{angle in alternate segment})$$

$$\therefore \angle BAP = \angle CAP$$

$$\therefore PA \text{ bisects } \angle BAC$$

$$b) \quad 2x^2 - 3x^2 - 2x + k = 0 \quad \text{triple}$$

$$8x^3 - 6x - 2 = 0 \quad \text{double}$$

$$24x^2 - 6 = 0 \quad \text{single}$$

solving

$$2x^2 = 6$$

$$x^2 = 3$$

substitute $\pm \sqrt{3}$ into 1st deriv

$$8(\pm \sqrt{3})^3 - 6(\pm \sqrt{3}) - 2 \neq 0 \quad \therefore \pm \sqrt{3} \text{ is not a root}$$

$$\therefore \text{triple root must be } x = \frac{1}{2}$$

sub into equation

$$2(\frac{1}{2})^2 - 3(\frac{1}{2})^2 - 2(\frac{1}{2}) + k = 0$$

Teacher's Name: _____ Student's Name/N^o: _____

Q.5

c) $x^4 - 6x^3 + 15x^2 - 18x + 10 = 0$

$1+i$ is a root

$1-i$ is a root

$(x-(1+i))(x-(1-i))$ is a factor

$x^2 - 2x + 2$ is a factor

$x^4 - 6x^3 + 15x^2 - 18x + 10 = 0$

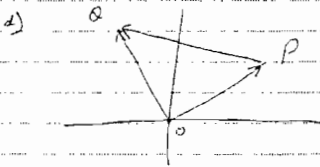
$(x^2 - 2x + 2)(x^2 - 4x + 5) = 0$

$x^2 - 4x + 5 = 0$

$x = \frac{4 \pm \sqrt{4^2 - 4 \cdot 1 \cdot 5}}{2}$

$= 2 \pm i$

∴ roots $2 \pm i, 1 \pm i$



i) $OA = a + ib$
 $= -b + ia$

ii) $PA = OA - OP$
 $= (-b + ia) - (a + ib)$
 $= -(a+b) + i(a-b)$

Teacher's Name: _____ Student's Name/N^o: _____

Q.6

a) $f(x) = e^{\frac{x}{1+kx}}$

$f(x) = e^{\frac{x}{1+kx}} \times \frac{(1+kx)' - (x)'(1+kx)'}{(1+kx)^2}$
 $= e^{\frac{x}{1+kx}} \times \frac{1 - (1+kx)}{(1+kx)^2}$

at $x=a$

$m = e^{\frac{a}{1+ka}} \times \frac{1}{(1+ka)^2}$

∴ equation of tangent is

$y - f(a) = \frac{f'(a)}{(1+ka)^2} (x-a)$

sub $(0,0)$

$-f(a) = e^{\frac{a}{1+ka}} \frac{1}{(1+ka)^2} (-a)$

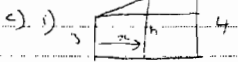
$-e^{\frac{a}{1+ka}} = -\frac{a e^{\frac{a}{1+ka}}}{(1+ka)^2}$

$1 = \frac{a}{(1+ka)^2}$

$k^2 a^2 + 2ka + 1 = a$

$k^2 a^2 + a(2k+1) = 0$

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∴ $h = \sqrt{4^2 - 3^2}$

when $x=0$ $h=3 \Rightarrow b=3$

at $x=4$ $h=0$

$\therefore 4 = 4a + 3$

$a = \frac{1}{4}$

$\therefore h = \frac{1}{4}x + 3$

ii) $A(x) = \frac{1}{2} \times \text{base} \times \text{height}$
 $= \frac{1}{2} \times 4x \times \left(\frac{1}{4}x + 3\right)$
 $= \frac{1}{2} \times 4x \times \sqrt{4x^2 + 24x + 9}$

$\therefore DV = \sqrt{4x^2 + 24x + 9} dx$

$V = \int_0^4 \sqrt{4x^2 + 24x + 9} dx$

$= \sqrt{4} \int_0^4 \frac{1}{2} x^2 + 3x + \frac{9}{4} dx$

$= \sqrt{4} \left[\frac{1}{6} x^3 + \frac{3}{2} x^2 + \frac{9}{4} x \right]_0^4$

$= \sqrt{4} \left[\left(\frac{2^3 \cdot 4}{3} + 3 \cdot 4 + \frac{9 \cdot 4}{4} \right) - (0) \right]$

$= \frac{96\sqrt{2}}{5}$ cu units

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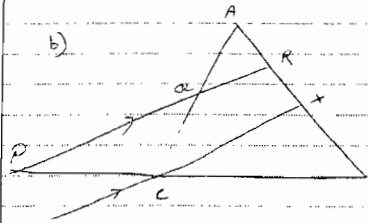
ii) no tangents if equation has no roots

i.e. $\Delta < 0$

$(2k+1)^2 - 4(1)(1) < 0$

$4k^2 - 4k + 1 - 4 < 0$

$4k^2 > 1$



$\frac{BP}{PC} = \frac{BR}{RQ}$ (intercepts in proportion)

$\frac{AQ}{QC} = \frac{AR}{RQ}$ (intercepts in proportion)

$RQ = \frac{AR \times QC}{AQ}$ sub into *

$\frac{BP}{PC} = \frac{BR}{\frac{AR \times QC}{AQ}}$

$\frac{BP \times AR \times QC}{AQ \times BR \times PC} = 1$

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Q7

$$i) I_1 = \int x^2 e^{2x} dx \quad u = x^2 \quad v = \frac{1}{2} e^{2x}$$

$$u' = 2x \quad v' = e^{2x}$$

$$\therefore I_1 = (x^2) \left(\frac{1}{2} e^{2x} \right) - \int (2x) \left(\frac{1}{2} e^{2x} \right) dx$$

$$= \frac{x^2 e^{2x}}{2} - \frac{1}{2} \int x e^{2x} dx$$

$$= \frac{x^2 e^{2x}}{2} - \frac{1}{2} I_2$$

$$ii) I_2 = \int x e^{2x} dx$$

$$= \frac{1}{2} x^2 e^{2x} - I_1$$

$$= \frac{1}{2} x^2 e^{2x} - \left[\frac{1}{2} x^2 e^{2x} - \frac{1}{2} I_1 \right]$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x^2 e^{2x} + \frac{1}{4} \int e^{2x} dx$$

$$= \frac{1}{4} x^2 e^{2x} - \frac{1}{4} x e^{2x} + \frac{1}{8} e^{2x} + C$$

$$= \frac{1}{8} e^{2x} (x^2 - 2x + 1) + C$$

b) $F = ma$

$$\therefore 2 \cdot a = -16 - v^2 \quad (-w \text{ is acting against motion})$$

$$\therefore a = -8 - \frac{1}{2} v^2$$

$$\frac{dv}{dt} = -8 - \frac{1}{2} v^2 \quad \text{or} \quad a = \frac{dv}{dt}$$

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$$i) \frac{dv}{dt} = -\left(8 + \frac{1}{2} v^2\right)$$

$$\frac{dv}{8 + \frac{1}{2} v^2} = -dt$$

$$\frac{2 dv}{16 + v^2} = -dt$$

$$\therefore \int \frac{2 dv}{16 + v^2} = -\int dt$$

$$\frac{1}{2} \tan^{-1} \frac{v}{4} = -t + C$$

$$\text{when } t=0, v=0$$

$$\therefore C = \frac{1}{2} \tan^{-1} \frac{0}{4}$$

$$\therefore t = \frac{1}{2} \tan^{-1} \frac{v}{4} - \frac{1}{2} \tan^{-1} \frac{0}{4}$$

$$2t = \tan^{-1} \frac{v}{4} - \tan^{-1} \frac{0}{4}$$

take $\tan$ of both sides

$$\tan 2t = \tan \left(\tan^{-1} \frac{v}{4} - \tan^{-1} \frac{0}{4} \right)$$

$$= \frac{\tan \tan^{-1} \frac{v}{4} - \tan \tan^{-1} \frac{0}{4}}{1 + (\tan \tan^{-1} \frac{v}{4})(\tan \tan^{-1} \frac{0}{4})}$$

$$= \frac{\frac{v}{4} - \frac{0}{4}}{1 + \frac{0v}{16}}$$

$$= \frac{v}{16 + 0v}$$

$$= \frac{4v}{16 + 0v}$$

$$2t = \tan^{-1} \left(\frac{4v}{16 + 0v} \right)$$

$$t = \frac{1}{2} \tan^{-1} \left(\frac{4v}{16 + 0v} \right)$$

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$$ii) V \frac{dv}{dx} = -8 - \frac{1}{2} v^2$$

$$\frac{V dv}{16 + v^2} = -\frac{1}{2} dx$$

Integrating

$$\int \frac{v dv}{16 + v^2} = -\frac{1}{2} \int dx$$

$$\frac{1}{2} \ln(16 + v^2) = -\frac{1}{2} x + C$$

$$\text{when } t=0, v=2 \quad x=20$$

$$\therefore C = \frac{1}{2} \ln(16 + 2^2)$$

$$\therefore x = \ln(16 + v^2) - \ln(16 + 4)$$

$$x = \ln \left(\frac{16 + v^2}{16 + 4} \right)$$

$$e^x = \frac{16 + v^2}{16 + 4}$$

$$e^{-x} = \frac{16 + 4}{16 + v^2}$$

$$16 + v^2 = (16 + 4) e^{-x}$$

$$v^2 = (16 + 4) e^{-x} - 16$$

or

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Q8

$$a) \sin(x + 10^\circ) = \cos 4x$$

$$\therefore \cos(90^\circ - (x + 10^\circ)) = \cos 4x$$

$$\therefore 90^\circ - (x + 10^\circ) = n \cdot 360^\circ \pm 4x$$

$$50^\circ - x = n \cdot 360^\circ + 4x$$

$$50^\circ - x = n \cdot 360^\circ - 50^\circ$$

$$x = \frac{1}{5} (4n \cdot 360^\circ - 50^\circ)$$

$$n=0 \quad x = 10^\circ$$

$$n=1 \quad x = 88^\circ$$

$$n=2 \quad x = 166^\circ$$

$$\therefore \text{Solutions: } 10^\circ, 88^\circ, 166^\circ$$

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b) Step 1 test $n=1$ $LHS = \tan \theta$

$$\begin{aligned}
 RHS &= \cot \theta - 2 \cot 2\theta \\
 &= \cot \theta - 2 \left(\frac{1 - \tan^2 \theta}{2 \tan \theta} \right) \\
 &= \cot \theta - \cot \theta + \tan \theta \\
 &= \tan \theta \\
 &= LHS
 \end{aligned}$$

 $\therefore$ true for $n=1$ Step 2 Assume true for $n=k$

$$i.e. \tan \theta + \tan 2\theta + \dots + 2^k \tan(2^k \theta) = \cot \theta - 2^k \cot(2^{k+1} \theta)$$

Step 3 show true for $n=k+1$

$$LHS = S_k + T_{k+1}$$

$$\begin{aligned}
 &= \cot \theta - 2^k \cot(2^{k+1} \theta) + 2^k \tan(2^k \theta) \\
 &= \cot \theta - 2^k \cot(2^{k+1} \theta) + 2^k (\cot(2^{k+1} \theta) - 2 \cot(2^{k+2} \theta)) \\
 &= \cot \theta - 2^{k+1} \cot(2^{k+2} \theta) \quad (* \text{ from above})
 \end{aligned}$$

 $\therefore$ true for $n=k+1$ if true for $n=k$

Step 4 As true for $n=1$ it is also true for $n=1+1$ i.e. $n=2$
 As true for $n=2$ it is also true for $n=2+1$ i.e. $n=3$
 and so on for all positive integers n .

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$$c) f(x) = 2 \log_e x - \frac{x^2-1}{x}$$

$$1) f(1) = 2 \log_e 1 - \frac{1^2-1}{1} = 0$$

 $\therefore$ given at $x=1$

$$\begin{aligned}
 f'(x) &= \frac{2}{x} - 1 - x^{-2} \\
 &= \frac{2x - x^2 - 1}{x^2} \\
 &= -\frac{(x^2 - 2x + 1)}{x^2} \\
 &= -\frac{(x-1)^2}{x^2}
 \end{aligned}$$

which is negative for all values of $x > 0$ except $x=1$ $\therefore$ curve always decreasing except at $x=1$ $\therefore$ can only be max given

$$ii) g(x) = \frac{x \log_e x}{x^2 - 1}$$

Consider $0 < x < 1$

$$x \log_e x < 0$$

$$x^2 - 1 < 0$$

$$\therefore g(x) > 0$$

$$x > 1$$

$$x \log_e x > 0$$

$$x^2 - 1 > 0$$

$$\therefore g(x) > 0$$

$$\therefore g(x) > 0 \text{ for all } x > 0, x \neq 1$$

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Q.8. a) ii) (cont.)

$$g(x) = \frac{x \log_e x}{x^2 - 1}$$

$$= \frac{x}{x^2 - 1} (\log_e x)$$

$$= \frac{x}{x^2 - 1} \cdot \frac{1}{2} (\ln x^2)$$

$$= \frac{x}{x^2 - 1} \times \frac{1}{2} \left[f(x) + \frac{x^2 - 1}{x} \right]$$

$$= \frac{1}{2} \left[1 + \frac{x f(x)}{x^2 - 1} \right]$$

for $0 < x < 1$

$$x^2 - 1 < 0$$

$$f(x) > 0 \text{ from part i)}$$

$$\therefore \frac{x f(x)}{x^2 - 1} < 0$$

$$g(x) < \frac{1}{2}$$

for $x > 1$

$$x^2 - 1 > 0$$

$$f(x) < 0 \text{ from part i)}$$

$$\therefore \frac{x f(x)}{x^2 - 1} < 0$$

$$\therefore g(x) < \frac{1}{2}$$

 $\therefore$ combining both results

$$0 < g(x) < \frac{1}{2}$$

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